

Use long division with exact answers

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Calculate $8,736 \div 24$. Copy or complete the first step.

2. Calculate $9,408 \div 32$.

3. Calculate $7,245 \div 35$.

4. Miro says: Miro subtracts the multiple correctly but forgets to bring down the next digit. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate $8,736 \div 24$. Copy or complete the first step.
Answer 364.
2. Calculate $9,408 \div 32$.
Answer 294.
3. Calculate $7,245 \div 35$.
Answer 207.
4. Miro says: Miro subtracts the multiple correctly but forgets to bring down the next digit. Is this correct? Explain.
Answer After each subtraction, bring down exactly one next place-value digit.

Use long division with exact answers

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Calculate $8,736 \div 24$.

6. Check this result using an inverse, estimate or known fact:
Calculate $7,245 \div 35$.

2. Calculate $9,408 \div 32$.

7. Calculate $9,984 \div 48$ and prove the answer by multiplication.

3. Calculate $7,245 \div 35$.

8. Identify and correct this misconception: Miro subtracts the multiple correctly but forgets to bring down the next digit.

4. Solve this independently and show every stage: Calculate $8,736 \div 24$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate $9,408 \div 32$.

10. Write and solve a similar question to: Calculate $7,245 \div 35$.

Teacher answers and checking prompts

1. Calculate $8,736 \div 24$.
Answer 364.
2. Calculate $9,408 \div 32$.
Answer 294.
3. Calculate $7,245 \div 35$.
Answer 207.
4. Solve this independently and show every stage: Calculate $8,736 \div 24$.
Answer 364.
5. Use a second method or representation for: Calculate $9,408 \div 32$.
Answer Expected result: 294. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Calculate $7,245 \div 35$.
Answer Expected result: 207. A valid check must be shown.
7. Calculate $9,984 \div 48$ and prove the answer by multiplication.
Answer 208 because $208 \times 48 = 9,984$.
8. Identify and correct this misconception: Miro subtracts the multiple correctly but forgets to bring down the next digit.
Answer After each subtraction, bring down exactly one next place-value digit.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Calculate $7,245 \div 35$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Use long division with exact answers

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Calculate $9,984 \div 48$ and prove the answer by multiplication.

6. Create a new example that tests the same idea as: Calculate $7,245 \div 35$.

2. Prove the answer to this question, rather than only calculating it: Calculate $8,736 \div 24$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate $9,408 \div 32$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 207.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro subtracts the multiple correctly but forgets to bring down the next digit.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Calculate $9,984 \div 48$ and prove the answer by multiplication.
Answer 208 because $208 \times 48 = 9,984$.
2. Prove the answer to this question, rather than only calculating it: Calculate $8,736 \div 24$.
Answer Expected result: 364. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate $9,408 \div 32$.
Answer Expected result: 294. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 207.
Answer One valid reconstruction is: Calculate $7,245 \div 35$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro subtracts the multiple correctly but forgets to bring down the next digit.
Answer After each subtraction, bring down exactly one next place-value digit.
6. Create a new example that tests the same idea as: Calculate $7,245 \div 35$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.